Statistical and Computational Models

GOVT 6039 · Cornell University · Fall 2026

Professor Daniel de Kadt

Lecture	Tue & Thu, 11:40am - 12:55pm (75 min)
Discussion section	Wed, 11:15am - 12:05pm (50 min)
Instructor	Daniel de Kadt
Office hours	By appointment · White Hall 206
TA	Georgy Tarasenko

Introduction

This is a graduate-level methods course in the Cornell Department of Government. It covers statistical and computational models beyond linear regression, taken from the perspective of statistical learning. We begin by recasting linear regression in that framework, then develop non-parametric and penalized regression. From there we work through likelihood and generalized linear models, along with Bayesian computation, and then the canonical supervised and unsupervised machine learning methods. The final third covers non-standard data — text, images, audio, and spatial.

The course

Preparation. There are no formal prerequisites, but the following are recommended: graduate-level probability, statistics, and regression, comfort with matrix notation and basic calculus, and enough **R** to load, clean, and model data. Familiarity with **Python** is not assumed – we will introduce it as the course progresses.

Your hours are divided as follows:

- **Lectures**, Tue & Thu, 75 minutes each. Lectures will be **device free** (laptops and phones).
- **Discussion section**, Wed, 50 minutes each, led by the TA. Bring a laptop.
- **Reading**, each lecture has required readings. There are some suggested readings too.
- **Software** is principally **R**, with some **Python** mixed in. You will also need to use **Quarto** and **Git**.

Books. We will rely on a few key texts, all of which are free. You may want to buy certain books to support the authors.

- **ISL** — James, Witten, Hastie & Tibshirani, *An Introduction to Statistical Learning*, 2nd ed. (2021). Free at https://www.statlearning.com/, in R and Python editions.
- **GLM** — Rodríguez, *Generalized Linear Models* (Princeton lecture notes). Free at https://grodri.github.io/glms/notes/.

- **BR** — Johnson, Ott & Dogucu, *Bayes Rules!* (2022). Free at https://www.bayesrulesbook.com/.

These books are also assigned multiple times, and are also free:

- **CASI** — Efron & Hastie, *Computer Age Statistical Inference* (2016). Free at https://hastie.su.domains/CASI/.
- **SLP** — Jurafsky & Martin, *Speech and Language Processing*, 3rd ed. draft. Free at https://web.stanford.edu/~jurafsky/slp3/.
- **GCR** — Lovelace, Nowosad & Muenchow, *Geocomputation with R*, 2nd ed. Free at https://r.geocompx.org/.
- **SDS** — Pebesma & Bivand, *Spatial Data Science* (2025). Free at https://r-spatial.org/book/.

Learning outcomes

By the end of the course, students will be able to:

1. Situate models within the statistical learning framework.
2. Derive, estimate, and interpret generalized linear and hierarchical models for binary, count, ordered, multinomial, duration, and grouped outcomes, by maximum likelihood.
3. Implement, tune, and evaluate supervised and unsupervised learning methods.
4. Convert text, images, audio, video, and spatial objects into data suitable for social scientific analysis.
5. Referee computational work.

Assessment

Component	Weight	When
Problem sets (4, bi-weekly)	0%	See calendar
Mid-term, in class	40%	**Tue Oct 27**
Reading presentation	10%	Assigned in week 2
Final project	40%	Paper due [DATE TBD]
Project presentation	10%	**Thur Dec 3**

Problem sets. There will be four ungraded problem sets. Each is released on a Wednesday and due two weeks later, also on a Wednesday (three weeks for the last, over Thanksgiving). You will receive feedback on your work, but they will not count toward your final grade. Collaboration is encouraged — attempt each problem alone first, write your own code, list your collaborators. Do not rely on AI to solve the problems or you won't learn anything.

Mid-term. There will be one in-class mid-term covering Units 1 and 2. Unit 3 is taught after the mid-term and is not examined on it. The mid-term uses the full 75-minute lecture. You may prepare two pages of hand-written or printed notes for the mid-term, but devices are strictly not allowed.

Reading presentation. You will each present one paper at some point throughout the semester. You should closely read the assigned paper and build a 10–12 minute long presentation, and expect about 10 minutes of Q&A and discussion.

Final project. Your final assignment is to develop an original project that applies a method from this course to an empirical problem of your choice. The final paper you deliver should be roughly **2,000–4,000 words** plus a **reproducible code repository**. Replication-and-extension is fine if the extension does something meaningfully new. Co-authoring with another student is permissible and encouraged.

Final project milestones	Date
Check-in meeting with the instructor	before **Tue Oct 6**
Presentations — 15 min plus peer review	**Thu Dec 3**
Paper and code due	**[DATE TBD]**

Calendar

The substance of the course is broken into four units. The dates below are subject to change without notice, depending on how the class progresses.

Problem sets are released and due on Wednesdays, listed below against the Tuesday of that week. Unit 3 is taught after the mid-term and is not examined on it.

Date	Topic	Reading	Due
	Unit 1 · Regression as statistical learning		
Tue Aug 25	Statistical and computational foundations	ISL 2.1; Spirling & Stewart 2025	
Thu Aug 27	Linear regression as statistical learning	ISL 2.2, 3.1–3.2	
Tue Sep 1	Nonparametric regression	ISL 3.5, 7.1–7.2	
Thu Sep 3	*APSA — no class*		
Tue Sep 8	Assessing models	ISL 5.1; Lundberg, Johnson & Stewart 2021	Papers assigned
Thu Sep 10	Penalization	ISL 6.1–6.2; CASI 7	
	Unit 2 · Likelihood, GLMs, and Bayesian computation		
Tue Sep 15	Maximum likelihood	CASI 4.1–4.2; GLM App. A	PS1 out
Thu Sep 17	Binary outcomes	GLM 3	
Tue Sep 22	Interpreting binary outcomes	Hanmer & Kalkan 2013	
Thu Sep 24	Interpreting binary outcomes, continued	—	
Tue Sep 29	Ordered and unordered outcomes	GLM 6; Bürkner & Vuorre 2019	PS1 due
Thu Oct 1	Count outcomes and comparing models	GLM 4	
Tue Oct 6	Catch-up: ordered, unordered, and count outcomes	—	Project check-in · PS2 out
Thu Oct 8	Duration and event history	ISL 11.1–11.4; Licht 2011	
Tue Oct 13	*Fall Break — no class*		
Thu Oct 15	Bayesian inference and the posterior	CASI 3; BR 6	
Tue Oct 20	Bayesian inference and the posterior, continued	—	PS2 due
Thu Oct 22	Hierarchical models and MRP	BR 18; Caughey & Warshaw 2019	
Tue Oct 27	**Mid-term, in class**	—	**Mid-term** · PS3 out
	Unit 3 · Learning beyond the linear model		
Thu Oct 29	Trees, random forests, and boosting	ISL 8.1–8.2; Montgomery & Olivella 2018	
Tue Nov 3	Splines, GAMs, and kernels	ISL 7.3–7.7; Hainmueller & Hazlett 2014	
Thu Nov 5	Neural networks and deep learning	ISL 10.1–10.2	
Tue Nov 10	Dimension reduction and latent variables	ISL 12.1–12.2.3, 12.4.1, 12.4.3; CASI 9.5; Grimmer & King 2011; Imai, Lo & Olmsted 2016	PS3 due · PS4 out
	Unit 4 · Non-standard data		
Thu Nov 12	Representing text as data	SLP 5.1–5.4, 2.4–2.5; Denny & Spirling 2018	
Tue Nov 17	Discovery, measurement, and embeddings	Grimmer, Roberts & Stewart 2021 sections 3–6; Ying, Montgomery & Stewart 2022; Rodriguez & Spirling 2022	
Thu Nov 19	Transformers and large language models	SLP 7.1–7.5, 8.1, 8.3–8.5; Gilardi et al. 2023; Egami et al. 2023	

Date	Topic	Reading	Due
Tue Nov 24	Images, video, and audio as data	SLP 15; Torres & Cantú 2022; Knox & Lucas 2021	
Thu Nov 26	*Thanksgiving — no class*		
Tue Dec 1	Spatial data and models	GCR 2; Neumayer & Plümper 2016	PS4 due
Thu Dec 3	**Project presentations**	—	**Presentations**

Course policies

Computer requirements

You need a computer that can run R, Python, Positron, Git and a C++ compiler. Full instructions, including what to install and in what order, are in `setup.md` on the course website.

You are responsible for having a working machine throughout the course. Please communicate early with the teaching staff early if this is a barrier.

Subject to change

All material, assignments, and dates are subject to change. It is your responsibility to stay in touch with the teaching staff and review the course site regularly.

Collaboration

Working together on problem sets is encouraged; talking through problems is how this material is learned. Attempt each problem alone first, write your own solutions and your own code, and list your collaborators. The mid-term is strictly individual. For the final project, co-authorship with another student is permitted and encouraged; co-authorship with faculty requires advance permission.

Grade disputes

To dispute a grade on the mid-term or an assignment: first, wait at least 48 hours after it is returned. Second, email me to set up an appointment, attaching a separate, typed summary of why you believe the grade is unfair. Revised grades may be higher or lower than the original, and the new grade is final.

Academic integrity

Each student in this course is expected to abide by the Cornell University Code of Academic Integrity. Any work submitted by a student in this course for academic credit will be the student's own work. Students are strictly forbidden from buying or selling course materials. Cheating and plagiarism will be treated according to Cornell University regulations, which I will strictly observe. Material drawn from research materials must be cited and, if a direct quote, demarked by quotation marks. The penalty for violating the Code of Academic Integrity is an F for the assignment or examination.

The Code is in effect throughout the semester. By taking this course, you affirm that it is a violation to cheat on the mid-term, to plagiarize, to deviate from the instructor's instructions about collaboration on graded work, to give false information to a faculty member, or to undertake any other form of academic misconduct.

Please review the Code and university policies on plagiarism at https://theuniversityfaculty.cornell.edu/dean/academic-integrity/code-of-academic-integrity/. If you have any doubts about what constitutes a violation, ask.

Artificial intelligence (AI) policy

This course is partly *about* large language models, and AI-assisted coding is now ordinary practice in computational social science. For this class the policies are as follows:

- **Mid-term:** no AI tools of any kind in completing the mid-term (you may use AI tools to prepare).
- **Problem sets and discussion section work:** generative AI is permitted as a coding and learning aid — debugging an error, explaining an unfamiliar function, translating an implementation from R to Python. It is not permitted as a substitute for doing the analytical work - if you do this you will be wasting your own time.
- **Final project:** generative AI is permitted but you should **disclose** in a short methods note how you used it. What you submit is your responsibility.

If you are unsure whether a particular use — or whether a particular tool counts as GAI — is permitted, ask before proceeding.

Expectations and attendance

You will get out of this course what you put into it. Regular attendance and active, prepared participation are expected in both the lecture and the discussion section. If an extraordinary or unavoidable circumstance will cause you to miss class, email the teaching staff in advance. (If you are sick, stay home — see *Mental and physical health* below.)

Devices in class

Your attention is your most valuable asset, and devices are designed to capture it. Bring a laptop to the discussion section (but not a mobile phone), but in lecture laptops and mobile phones should be turned off or put away. If you have any concerns about this policy please reach out directly to me (not the TA).

Use of recording devices

Students must request permission from the instructor before making any audio or video recording of lectures, discussion sections, or presentations. If permission is granted, recordings and any transcripts or copies may be used only as a replacement for personal notes and may not be shared or distributed without explicit permission. This is to create a learning environment where students can express their opinions without the possibility of their contribution being reproduced, distributed, and replayed without their consent. Permission will not be withheld where a recording is a reasonable accommodation for a student's disability.

Accommodations for students with disabilities (SDS)

Your access to this course is important. In compliance with Cornell University policy and equal access laws, I am available to discuss appropriate academic accommodations. Please request your accommodation letter early in the semester, or as soon as you register with Student Disability Services (SDS), so that we have adequate time to arrange approved accommodations; letters should be submitted within the first two weeks of the semester where possible. Once SDS approves your letter, it will be emailed to you and the instructor; please follow up to discuss logistics. If you experience any access barriers in this course — printed content, graphics, online materials, or

communication — reach out to the instructor or your SDS counselor right away. Contact SDS as early as possible: https://sds.cornell.edu · 607-254-4545.

Religious observances

Students may ask for reasonable and timely accommodations for sincerely held religious beliefs. Please review the calendar above closely to determine whether your religious observance will present any scheduling conflicts with lectures, discussion sections, the mid-term, or project milestones. Please inform me as early as possible about any clashes.

Inclusion and diversity

It is the instructor's intent that students from all backgrounds and perspectives be well served by this course, that students' learning needs be addressed both in and out of class, and that the diversity students bring to the class be viewed as a resource, strength, and benefit. The instructor aims to present materials and activities that are respectful of diversity along lines of gender identity, sexuality, disability, age, socioeconomic status, ethnicity, race, nationality, religion, and culture. I affirm people of all gender expressions and gender identities; if you prefer to be called a different name than the one indicated on the class roster, or wish to correct me on your pronouns, please let me know. Suggestions for improving the course for you or for other students are welcome at any time.

Classroom behavior

Students and faculty each have responsibility for maintaining an appropriate learning environment. Students will treat one another with respect and courtesy. Disruptive students will be asked to leave the class.

Mental and physical health

Learning happens best when you are physically and mentally healthy. If you find yourself experiencing distress that affects your studies, please consult Cornell Health. Counselors are available 24/7 at 607-255-5155, and additional support (including Let's Talk) is described at https://health.cornell.edu.

Coming to class sick. It is best for you — and for everyone else in the room — that you do not come to class when you are sick. **You will not be penalized for staying home when you are sick**, provided that you notify the teaching staff as soon as possible. You are ultimately responsible for catching up on material you missed by reviewing the posted materials and getting notes from a peer. Wearing a mask is always available to anyone who chooses it, and if you must attend class while sick, or if you're unsure whether you're sick, please wear one.

University resources

This course aims to collectively create a welcoming, supportive and tolerant environment for all students and respects the various forms of diversity that they bring to the classroom. The university is committed to full inclusion in education for all persons. Services and reasonable accommodations are available to persons with temporary and permanent disabilities, students with DACA or undocumented status, students facing mental health or other personal challenges, and students

with other kinds of learning challenges. Please feel free to let me know if there are circumstances affecting your ability to participate in class.

- Office of Student Disability Services: https://sds.cornell.edu
- Cornell Health CAPS (Counseling & Psychological Services): https://health.cornell.edu/services/counseling-psychiatry
- Undocumented/DACA student support: https://dos.cornell.edu/undocumented-daca-support/undergraduate-admissions-financialaid
- Additional academic support is available through the Learning Strategies Center, the Knight Institute for Writing in the Disciplines, and the Office of Academic Diversity Initiatives.

I respect and uphold Cornell University policies and regulations pertaining to the observation of religious holidays; assistance available to students with disabilities and to visually and/or hearing impaired students; sexual harassment; and racial or ethnic discrimination.

Cayuga Nation acknowledgment

Cornell University is located on the traditional homelands of the Gayogo̲hó:nǫ' (the Cayuga Nation). The Gayogo̲hó:nǫ' are members of the Haudenosaunee Confederacy, an alliance of six sovereign nations with a historic and contemporary presence on this land. The confederacy precedes the establishment of Cornell University, New York State, and the United States of America. We acknowledge the painful history of Gayogo̲hó:nǫ' dispossession, and honor the ongoing connection of Gayogo̲hó:nǫ' people, past and present, to these lands and waters.

Acknowledgments

Teaching — like scientific inquiry — is about standing on the shoulders of giants. The content of this course has been influenced and adapted from materials developed by Jens Hainmueller, Chad Hazlett, Danny Hidalgo, Ryan Hübert, Melissa Sands, Teppei Yamamoto, and an army of TAs at various institutions.